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(54) **OPTOELECTRONIC DEVICE INCLUDING LATERALLY DISPOSED DRIVING ELECTRODE**

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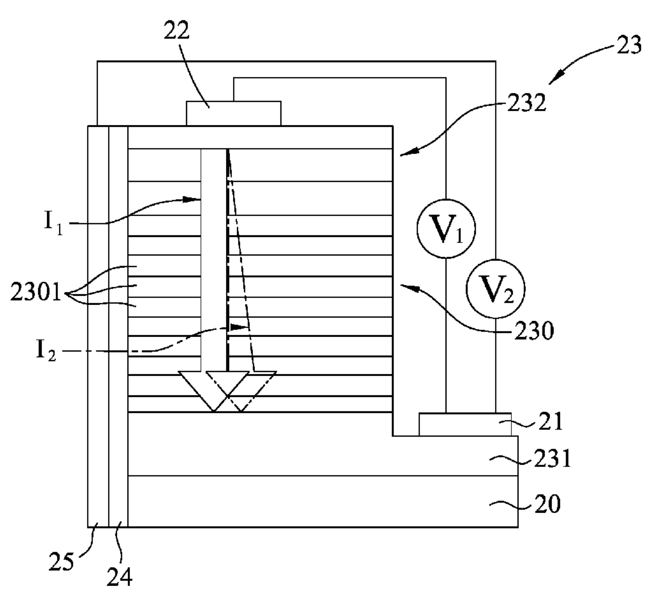
(57) **ABSTRACT**

An optoelectronic device includes a first electrode, a second electrode that is spaced apart from the first electrode, an optoelectronic unit that is disposed between the first electrode and the second electrode, an insulating layer and a driving electrode. The optoelectronic unit includes an optoelectronic stack emitting or absorbing at least two wavelengths of light. The insulating layer is disposed on a lateral side of the optoelectronic stack that extends in a stacking direction of the optoelectronic stack. The driving electrode is disposed on the insulating layer at a location corresponding in position to the optoelectronic unit and is separated from the first and second electrodes.

4 Claims, 8 Drawing Sheets



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H10F 77/20 (2025.01)
H10H 20/812 (2025.01)
H10H 20/831 (2025.01)

(58) **Field of Classification Search**
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See application file for complete search history.

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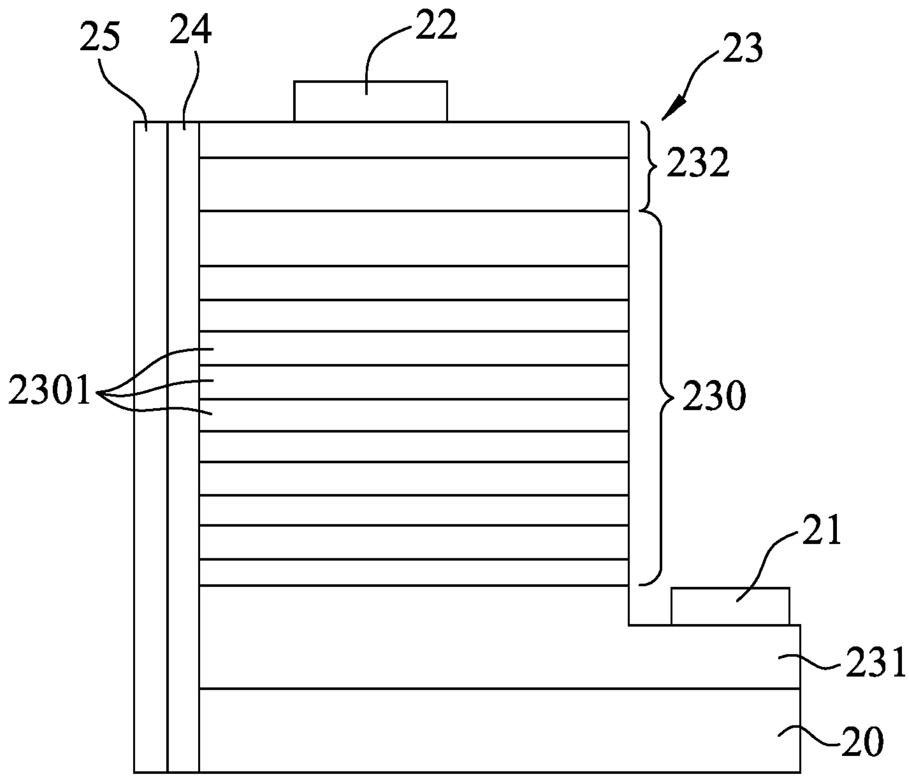

FIG.1

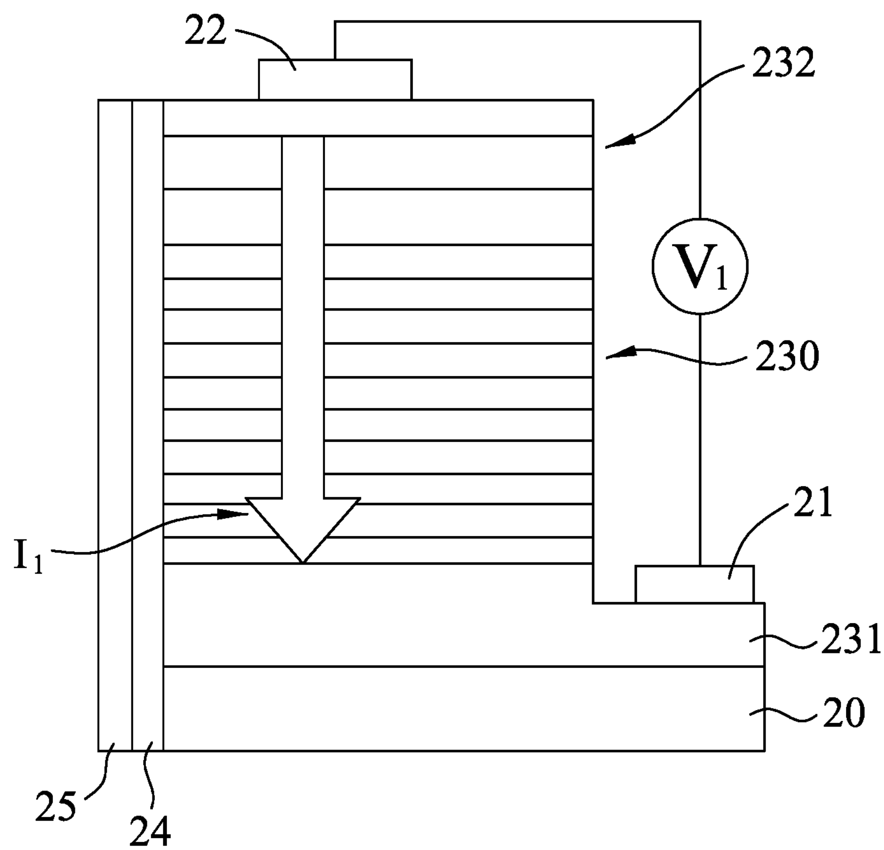

FIG.2

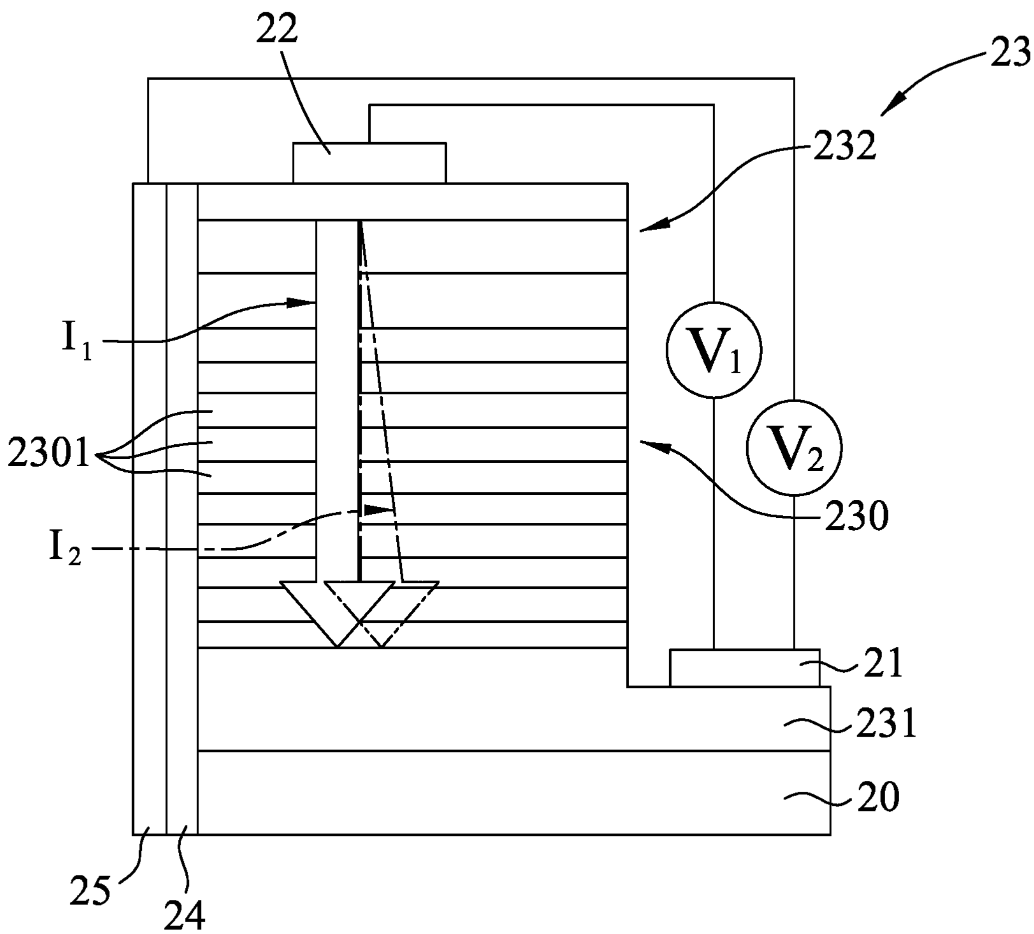

FIG.3

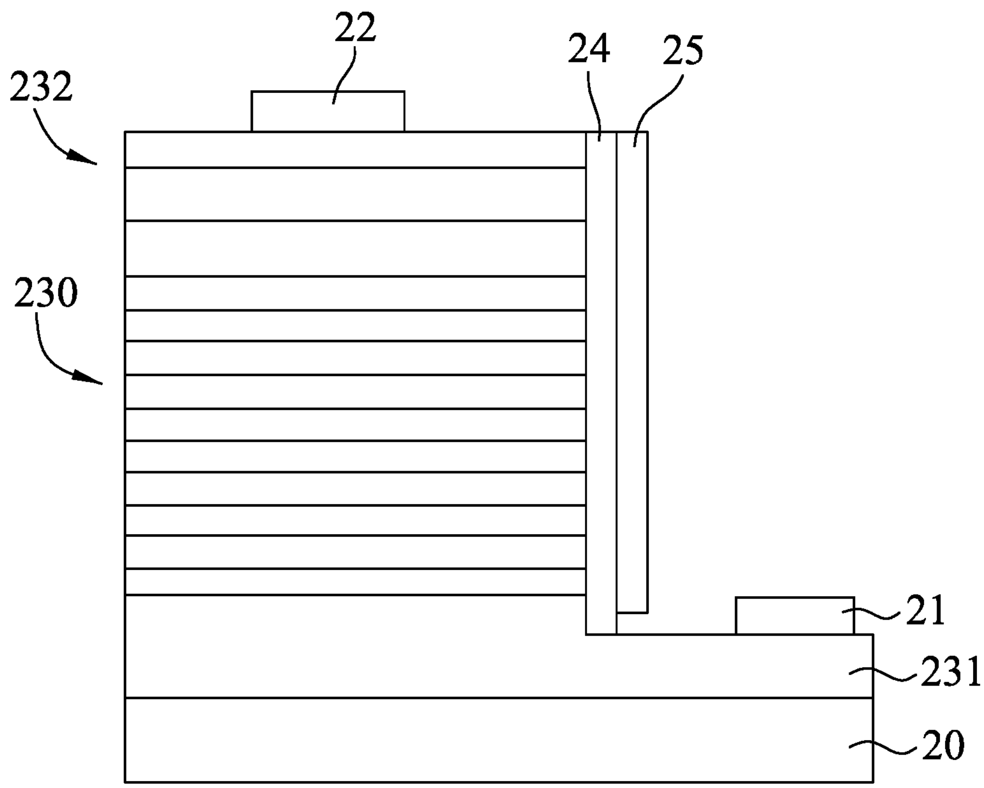

FIG.4

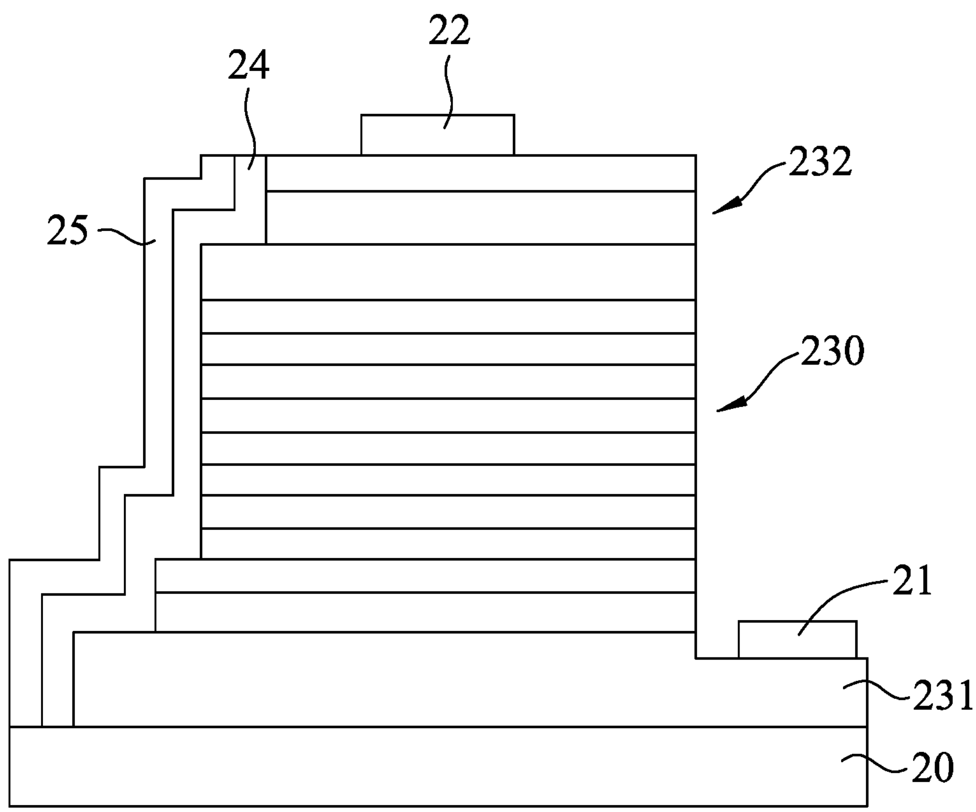

FIG.5

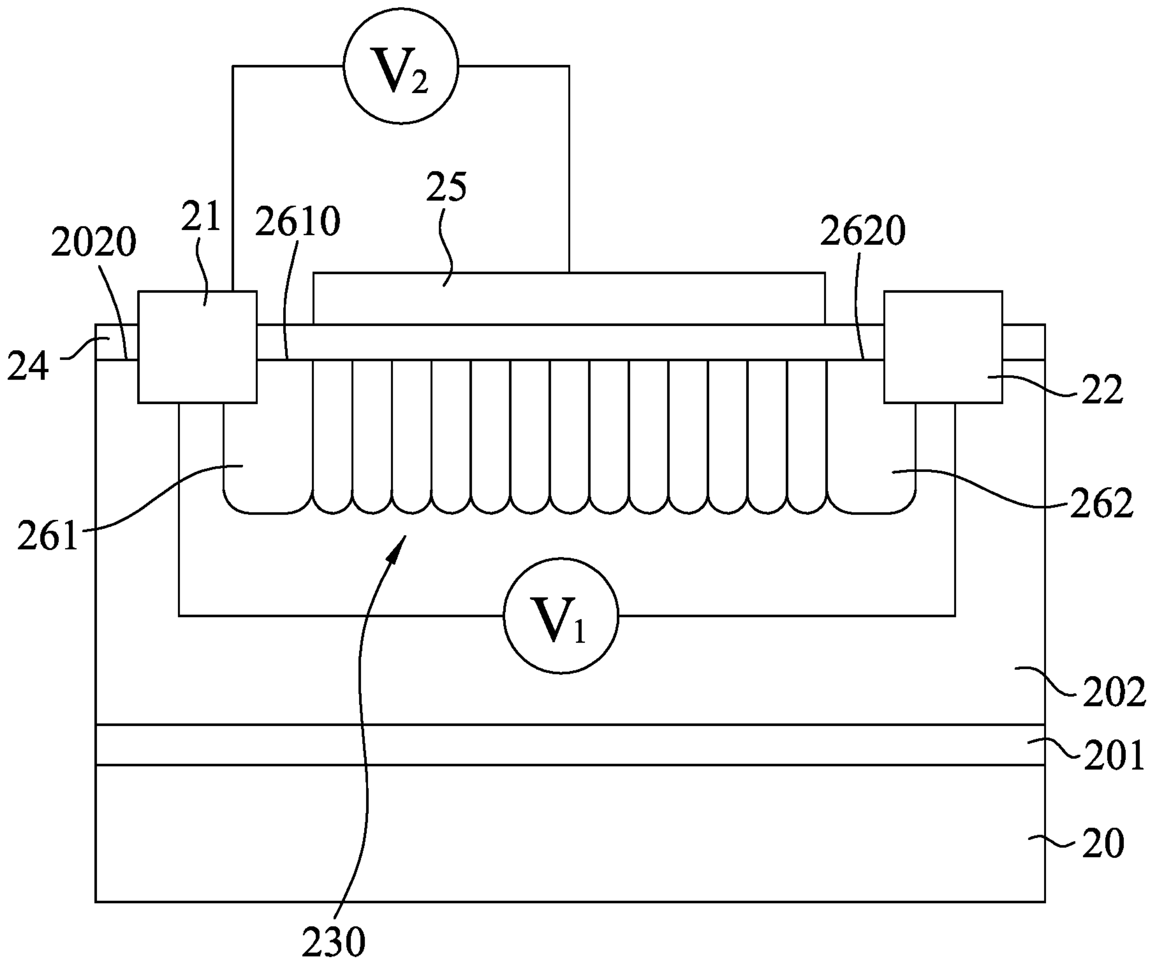

FIG.6

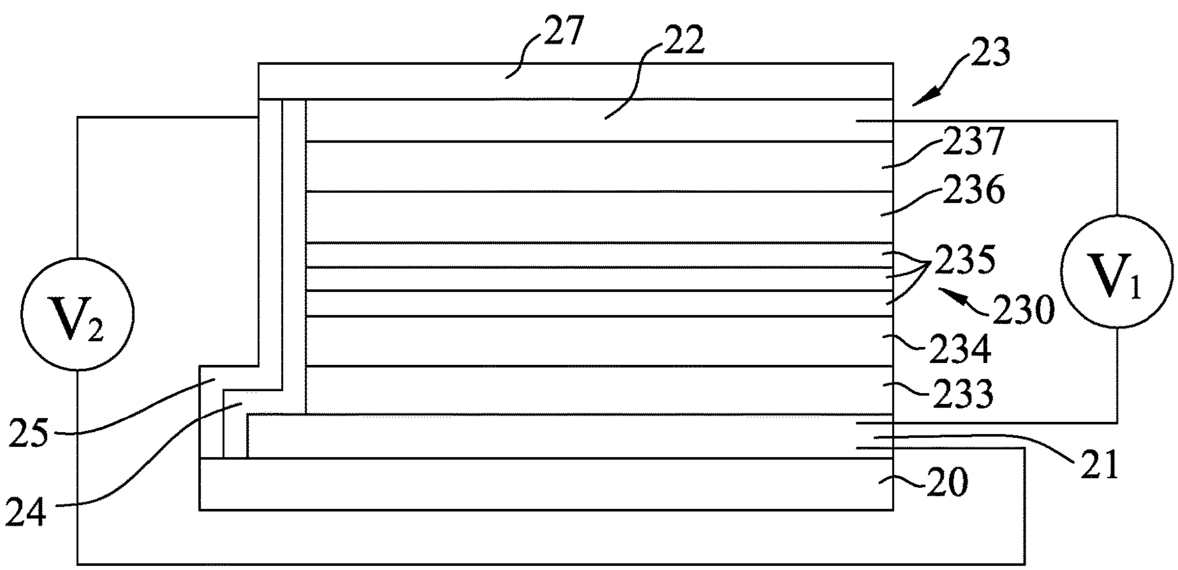

FIG.7

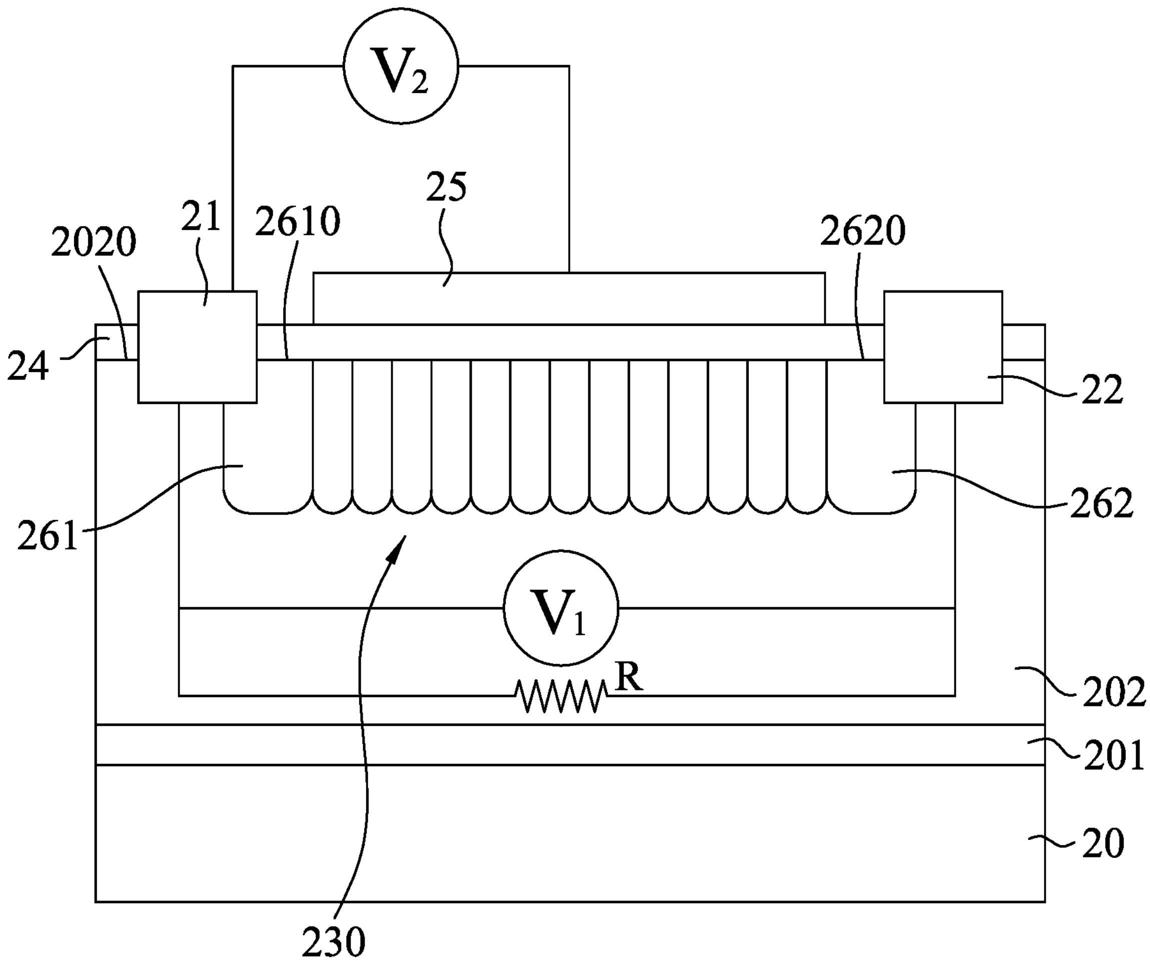

FIG.8

OPTOELECTRONIC DEVICE INCLUDING LATERALLY DISPOSED DRIVING ELECTRODE

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority to Taiwanese Invention Patent Application No. 111136090, filed on Sep. 23, 2022.

FIELD

The disclosure relates to an optoelectronic device, and more particularly to a color adjustable light emitting device and a solar cell.

BACKGROUND

In a light emitting diode (LED), the color of a light emitted from the LED is determined by the band gap of the semiconductor used in the LED. Generally, a conventional LED only emits one color of light (i.e., a color-specific LED) and the color of the LED may not be adjusted. When a desired or predetermined color of light cannot be obtained from one LED, a plurality of color-specific LEDs have to be used to mix the colors of light emitted therefrom to produce the desired or predetermined color of light. This increases operational difficulties and lacks operational flexibility.

SUMMARY

Therefore, an object of the disclosure is to provide an optoelectronic device that can alleviate at least one of the drawbacks of the prior art.

According to the disclosure, the optoelectronic device includes a first electrode, a second electrode that is spaced apart from the first electrode, an optoelectronic unit that is disposed between the first electrode and the second electrode, an insulating layer and a driving electrode. The optoelectronic unit includes an optoelectronic stack emitting or absorbing at least two wavelengths of light. The insulating layer is disposed on a lateral side of the optoelectronic stack that extends in a stacking direction of the optoelectronic stack. The driving electrode is disposed on the insulating layer at a location corresponding in position to the optoelectronic unit and is separated from the first and second electrodes.

BRIEF DESCRIPTION OF THE DRAWINGS

Other features and advantages of the disclosure will become apparent in the following detailed description of the embodiment(s) with reference to the accompanying drawings. It is noted that various features may not be drawn to scale.

FIG. **1** is a cross-sectional view illustrating a first embodiment of an optoelectronic device according to the disclosure.

FIG. **2** is a cross-sectional view illustrating a first voltage applying unit used in the first embodiment.

FIG. **3** is a cross-sectional view illustrating first and second voltage applying units used in the first embodiment.

FIG. **4** is a cross-sectional view illustrating a first variation of the first embodiment.

FIG. **5** is a cross-sectional view illustrating a second variation of the first embodiment.

FIG. **6** is a cross-sectional view illustrating a second embodiment of an optoelectronic device according to the disclosure.

FIG. **7** is a cross-sectional view illustrating a third embodiment of an optoelectronic device according to the disclosure.

FIG. **8** is a cross-sectional view illustrating a fourth embodiment of an optoelectronic device according to the disclosure.

DETAILED DESCRIPTION

Before the disclosure is described in greater detail, it should be noted that where considered appropriate, reference numerals or terminal portions of reference numerals have been repeated among the figures to indicate corresponding or analogous elements, which may optionally have similar characteristics.

Referring to FIG. **1**, a first embodiment of an optoelectronic device according to the disclosure includes a first electrode **21**, a second electrode **22** that is spaced apart from the first electrode **21**, an optoelectronic unit **23** that is disposed between the first electrode **21** and the second electrode **22**, an insulating layer **24** and a driving electrode **25**. The optoelectronic unit **23** includes an optoelectronic stack **230** capable of emitting at least two wavelengths of light. The insulating layer **24** is disposed on and covers a lateral side of the optoelectronic stack **230** that extends in a stacking direction of the optoelectronic stack **230**. The driving electrode **25** is disposed on the insulating layer **24** at a location corresponding in position to the optoelectronic unit **23** and is separated from the first and second electrodes **21**, **22**.

In this embodiment, the optoelectronic device is an inorganic light emitting device and may be a color adjustable light emitting device, and the optoelectronic stack **230** is a light emitting stack. The optoelectronic device further includes a substrate **20**, and the optoelectronic unit **23** further includes a first-type epitaxial layer **231** and a second-type epitaxial layer **232**.

To be specific, the first-type epitaxial layer **231** is disposed on the substrate **20**, and the optoelectronic stack **230** is disposed between the first-type epitaxial layer **231** and the second-type epitaxial layer **232**. The first electrode **21** is disposed on and contacts the first-type epitaxial layer **231**, and the second electrode **22** is disposed on and contacts the second-type epitaxial layer **232**.

In this embodiment, a portion of the first-type epitaxial layer **231** opposite to the substrate **20** is exposed, and the first electrode **21** is directly disposed on the exposed portion of the first-type epitaxial layer **231**. The insulating layer **24** covers a lateral side of the first-type epitaxial layer **231**, a lateral side of the optoelectronic stack **230** and a lateral side of the second-type epitaxial layer **232** that are opposite to the first electrode **21**, and the driving electrode **25** completely covers the insulating layer **24**.

It should be noted that the first-type semiconductor layer **231** is one of a p-type semiconductor layer and an n-type semiconductor layer, and the second-type semiconductor layer **232** is the other one of a p-type semiconductor layer and an n-type semiconductor layer. In this embodiment, the first-type semiconductor layer **231** is an n-type semiconductor layer and the second-type semiconductor layer **232** is a p-type semiconductor layer. The optoelectronic stack **230** is formed in a multiple quantum well (MQW) structure and includes a plurality of stack units **2301** each of which is constituted of an $In_xGa_{1-x}N$ film layer and a GaN film layer.

For example, the optoelectronic stack **230** includes eleven stack units **2301**, i.e., twenty two film layers. To be specific, the eleven stack units **2301** include two stack units of $In_{0.2}Ga_{0.8}N$/GaN (four film layers), two stack units of $In_{0.23}Ga_{0.77}N$/GaN, two stack units of $In_{0.26}Ga_{0.74}N$/GaN (four film layers), one stack unit of $In_{0.3}Ga_{0.7}N$/GaN (two film layers), one stack unit of $In_{0.33}Ga_{0.67}N$/GaN (two film layers), one stack unit of $In_{0.35}Ga_{0.65}N$/GaN (two film layers), one stack unit of $In_{0.4}Ga_{0.6}N$/GaN (two film layers) and one stack unit of $In_{0.5}Ga_{0.5}N$/GaN (two film layers), all of which are stacked sequentially one on another on the first-type semiconductor layer **231** in such an order. The colors of lights emitted from the stack units **2301** may be adjusted by changing the amounts of In and Ga in the stack units **2301** during the epitaxial process so that the optoelectronic stack **230** can produce a reddish light, a greenish light and a bluish light. It should be noted that the materials of each of the stack unit **2301** are not limited to $In_xGa_{1-x}N$ and GaN, and the number of the stack units are not limited.

In this embodiment, the first-type semiconductor layer **231** is made of n-GaN, and the second-type semiconductor layer **232** is a two-layer structure that includes, but not limited to, a layer of p-$Al_{0.1}Ga_{0.9}N$ formed on the optoelectronic unit **230** and a layer of p-GaN formed on the p-$Al_{0.1}Ga_{0.9}N$ layer.

Referring to FIGS. **2** and **3**, the first electrode **21** functions as a cathode, and the second electrode **22** functions as an anode. The optoelectronic device further includes a first voltage applying unit (V_1) that is adapted to apply a first voltage and generate a first electric field between the first and second electrodes **21**, **22** (see FIGS. **2** and **3**), and a second voltage applying unit (V_2) that is adapted to apply a second voltage and generate a second electric field between the first electrode **21** and the driving electrode **25** (see FIG. **3**). The first voltage produces a first current (I_1) passing through the optoelectronic unit **23**, i.e., the second-type epitaxial layer **232**, the optoelectronic stack **230** and the first-type epitaxial layer **231**. The second voltage produces a second current (I_2) that is different from the first current (I_1). The flow directions of the first current (I_1) and the second current (I_2) are influenced by relative positions of the first electrode **21**, the second electrode **22** and the driving electrode **25** and may be different from each other.

Referring to FIG. **2**, when the first voltage is applied between the first and second electrodes **21**, **22**, an overall current is contributed only by the first current (I_1) and is uniformly distributed over the optoelectronic stack **230** between the first and second electrodes **21**, **22**. Therefore, the optoelectronic stack **230** has a uniform current density. Referring to FIG. **3**, when the second voltage is additionally applied between the first electrode **21** and the driving electrode **25**, the overall current is contributed by the first current (I_1) and the second current (I_2) and is not uniformly distributed over the optoelectronic stack **230** between the first and second electrodes **21**, **22**. Therefore, the optoelectronic stack **230** has a non-uniform current density. To be specific, an upper portion of the optoelectronic stack **230** adjacent to the second-type epitaxial layer **232** has a current density greater than that of a lower portion of the optoelectronic stack **230** adjacent to the first-type epitaxial layer **231**.

The optoelectronic stack **230** can emit different wavelengths of light because the multiple stack units **2301** of the optoelectronic stack **230** can respectively emit different wavelengths of light (i.e., different light colors). As the driven electrode **25** cooperates with the first and second electrodes **21**, **22**, the overall current flowing through the optoelectronic stack **230** is changeable by controlling the first and second currents (I_1, I_2). In this embodiment, the second voltage applying unit (V_2) is controllable to adjust the second current (I_2) relative to the first current (I_1), as well as adjust a current density difference occurring in the optoelectronic stack **230**. Thus, the overall current distribution in the optoelectronic stack **230** may be controlled and the wavelengths of light emitted by the optoelectronic stack **230** may be adjusted.

In FIG. **2**, the optoelectronic stack **230** can emit a red light, a green light and a blue light when the first voltage applying unit (V_1) applies the first voltage between the first and second electrodes **21**, **22** to create the first current (I_1). The overall color of light emitted from the optoelectronic device is a mixture of the colors of lights emitted from the stack units **2301** of the optoelectronic stack **230**. However, the overall color of light emitted from the optoelectronic stack **230** cannot be varied or adjusted, because the different colors of light are emitted from equal areas of the materials of the stack units **2301**, that is to say, the different colors of light are emitted from equal light emission areas. In this case, the colors of light emitted respectively from the stack units **2301** are in a ratio of red light, green light and blue light which is constant and not adjustable.

In FIG. **3**, when the first and second voltages are applied, the first and second currents (I_1, I_2) flow between the first and second electrodes **21**, **22** and pass through the optoelectronic stack **230**. Under this condition, the overall current (i.e. the first and second currents (I_1, I_2)) is not distributed evenly in the optoelectronic stack **230** so that effective light emission areas of the light emitting stack units **2301** differ from each other, and the colors emitted respectively from the different stack units **2301** will have different light intensities. Since the intensities of red light, green light and blue light emitted from the optoelectronic stack **230** may be varied through a control of the difference in the current densities of the stack units **2301**, the color of light emitted from the optoelectronic device may be adjusted.

For example, when the second voltage applying unit (V_2) applies a small second voltage between the driving electrode **25** and the first electrode **21**, the current densities in the stack units **2301** differ from each other; but the magnitude of current density differences occurring in the stack units **2301** are small. In this case, the light emission areas that emit the red and green colors are medium size areas, and the light emission area that emits the blue color is a large size area so that the intensities of red and green colors of lights are lower than that of the blue color. When the second voltage applied between the driving electrode **25** and first electrode **21** becomes large, the magnitude of current density differences occurring in the stack units **2301** are large. In this case, the light emission areas that emit the red and green colors are small size areas, and the light emission area that emits the blue color is a large size area so that the intensities of the red and green colors of lights are much lower than that of the blue color.

It should be noted that the spatial arrangement of the first electrode **21**, the second electrode **22** and the driving electrode **25** is not limited to a vertically or horizontally spaced apart orientation relative to each other, and the optoelectronic unit **23** may be formed in any shape or structural arrangement.

Referring FIG. **4**, which shows a first variation of the first embodiment, the insulating layer **24** along with the driving electrode **25** may be disposed on a lateral side of the optoelectronic stack **230** that is adjacent to the first electrode **21**, and the driving electrode **25** is separated from the first electrode **21**. Referring to FIG. **5**, which shows a second variation of the first embodiment, the optoelectronic unit **23** has one lateral side in a stepped form, and the insulating layer **24** along with the driving electrode **25** are disposed on the lateral side of the optoelectronic unit **23**. Alternatively, the optoelectronic unit **23** may have two lateral sides both in irregular forms.

Referring to FIG. **6**, a second embodiment of the optoelectronic device according to the disclosure includes the substrate **20**, the first electrode **21**, the second electrode **22**, the optoelectronic unit **23**, the insulating layer **24** and the driving electrode **25** as illustrated in the first embodiment, and further includes a buffer layer **201** that is disposed on top of the substrate **20**.

In this embodiment, the optoelectronic unit **23** further includes an epitaxial layer **202**, a first-type semiconductor layer **261** and a second-type semiconductor layer **262**. The epitaxial layer **202** is disposed on top of the buffer layer **201** opposite to the substrate **20**. The first-type and second-type semiconductor layers **261**, **262** are formed in the epitaxial layer **202** in an embedded manner and spaced apart from each other in a direction transverse to a stacking direction of the epitaxial layer **202**. The first-type and second-type semiconductor layers **261**, **262** has outer surfaces **2610**, **2620** exposed from a top surface **2020** of the epitaxial layer **202**.

The optoelectronic stack **230** is disposed in the epitaxial layer **202** and is sandwiched between the first-type semiconductor layer **261** and the second-type semiconductor layer **262**. The stacking direction of the first-type and second-type semiconductor layers **261**, **262** and the optoelectronic stack **230** is transverse to the stacking direction of the epitaxial layer **202**. The lateral side of the optoelectronic stack **230** is exposed from the top surface **2020** of the epitaxial layer **202**. The insulating layer **24** covers the top surface **2020** of the epitaxial layer **202**, the outer surfaces **2610**, **2620** of the first-type semiconductor layer **261** and the second-type semiconductor layers **262**, and the lateral side of the optoelectronic stack **230**. The first electrode **21** and the second electrode **22** respectively contact the first-type and said second-type semiconductor layers **261**, **262** at locations distal from the optoelectronic stack **230** and partially protruding outward from the epitaxial layer **202** through the insulating layer **24**. In this embodiment, the first-type semiconductor layer **261** is an n-type semiconductor layer and the second-type semiconductor layer **262** is a p-type semiconductor layer.

Referring to FIG. **7**, a third embodiment of the optoelectronic device according to the disclosure is similar to the first embodiment, except that the third embodiment is an organic light emitting diode (OLED) device and the optoelectronic unit **23** has a different configuration.

In this embodiment, the optoelectronic device further includes a protective layer **27**. The first electrode **21** is disposed on the substrate **20**. The optoelectronic unit **23** further includes an electron injecting layer **233**, an electron transferring layer **234**, a plurality of light emitting layers **235**, a hole transferring layer **236** and a hole injecting layer **237** disposed on the first electrode **21** opposite to the substrate **20** in such order. The second electrode **22** is disposed on the hole injecting layer **237** opposite to the hole transferring layer **236**. The protective layer **27** is disposed on the second electrode **22** opposite to the hole injecting layer **237**. The light emitting layers **235** form the optoelectronic stack **230**. The insulating layer **24** and the driving electrode **25** are disposed in a same manner as in the first embodiment.

In this embodiment, the light emitting layers **235** emit different colors of light and have the same function as the light emitting layers **2301** in the first embodiment. It should be noted that the material used to form each layer of the optoelectronic device (i.e. OLED device) is well known by a person skilled in the art and thus is not illustrated herein.

The optoelectronic devices of the first, second and third embodiments, which are color adjustable light emitting devices, may be applied to a semiconductor material processing apparatus. In such application, the optoelectronic device may emit an adjustable color of light having a desired wavelength that may be absorbed by the material to be processed. For example, when the optoelectronic device is to be used for processing two different materials, such as silicon and gallium arsenide, it needs to emit two different wavelengths of light for the materials to absorb, respectively. For instance, under the same absorbing coefficient, silicon may absorb the light of 0.4 um wavelength whereas gallium arsenide may absorb the light of 0.6 um wavelength. Because the optoelectronic device can adjust the wavelength of light emitted therefrom so as to match with the wavelength to be absorbed by the material under processing, only one processing apparatus is necessary for processing one or more semiconductor materials. Thus, the optoelectronic device can solve the disadvantages of the prior art in which a single one processing apparatus can only be used for one wavelength of light, and two or more processing apparatuses are necessary to respectively process two or more semiconductor materials, thereby reducing the cost of fabrication and time.

Referring to FIG. **8**, a fourth embodiment of the disclosure which has a structure generally similar to that of the second embodiment. However, this embodiment is used as a solar cell which can absorb a wide range of wavelengths. The first electrode **21** is an anode and the second electrode **22** is a cathode, and the optoelectronic stack **230** absorbs different wavelengths of light, and generates and stores electrons and holes. By generating the first electric field between the first and second electrodes **21**, **22** and generating the second electric field between the first electrode **21** and the driving electrode **25**, the electrons and holes may be effectively separated and stored in the optoelectronic device. To be specific, the electrons and holes may be stored in an external load (i.e., a resistor (R)) that is connected between the first and second electrodes **21**, **22** and is connected with the first voltage supplying unit (V_1) in parallel.

To sum up, when the optoelectronic device is used as a color adjustable light emitting device, the optoelectronic stack **230** emits multiple wavelengths of light, and the two voltage applying units (V_1, V_2) enable the optoelectronic device to adjust the current density difference occurring in the optoelectronic stack **230** and thereby adjust the ratio of lights having different wavelengths (i.e., the colors of light). When the optoelectronic device is used as a solar cell, the optoelectronic stack **230** absorbs multiple wavelengths of light. The first and second voltage applying units (V_1, V_2) enable the optoelectronic device to separate the generated electrons and holes more effectively and increase the conversion efficiency.

In the description above, for the purposes of explanation, numerous specific details have been set forth in order to provide a thorough understanding of the embodiment(s). It will be apparent, however, to one skilled in the art, that one or more other embodiments may be practiced without some of these specific details. It should also be appreciated that reference throughout this specification to "one embodiment," "an embodiment," an embodiment with an indication of an ordinal number and so forth means that a particular feature, structure, or characteristic may be included in the practice of the disclosure. It should be further appreciated that in the description, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of various inventive aspects; such does not mean that every one of these features needs to be practiced with the presence of all the other features. In other words, in any described embodiment, when implementation of one or more features or specific details does not affect implementation of another one or more features or specific details, said one or more features may be singled out and practiced alone without said another one or more features or specific details. It should be further noted that one or more features or specific details from one embodiment may be practiced together with one or more features or specific details from another embodiment, where appropriate, in the practice of the disclosure.

While the disclosure has been described in connection with what is(are) considered the exemplary embodiment(s), it is understood that this disclosure is not limited to the disclosed embodiment(s) but is intended to cover various arrangements included within the spirit and scope of the broadest interpretation so as to encompass all such modifications and equivalent arrangements.

What is claimed is:

1. An optoelectronic device, comprising:

a first electrode;

a second electrode that is spaced apart from said first electrode;

an optoelectronic unit that is disposed between said first electrode and said second electrode and includes an optoelectronic stack emitting or absorbing at least two wavelengths of light;

an insulating layer that is disposed on a lateral side of said optoelectronic stack that extends in a stacking direction of said optoelectronic stack;

a driving electrode that is disposed on said insulating layer at a location corresponding in position to said optoelectronic unit and that is separated from and without being in contact with said first electrode and said second electrode; and

a first voltage applying unit that is adapted to apply a first voltage between said first electrode and said second electrode, and a second voltage applying unit that is adapted to apply a second voltage between said first electrode and said driving electrode, said first voltage producing a first current passing through said optoelectronic unit, said second voltage producing a second current that is different in flow direction from said first current.

2. The optoelectronic device as claimed in claim **1**, wherein said second voltage applying unit is controllable to adjust a current density difference occurring in said optoelectronic stack and thereby adjust the wavelengths of light.

3. The optoelectronic device as claimed in claim **1**, wherein said optoelectronic device is a color adjustable light emitting device, said first electrode functions as a cathode, and said second electrode functions as an anode.

4. The optoelectronic device as claimed in claim **1**, further comprising a substrate, wherein said optoelectronic unit further includes a first-type epitaxial layer and a second-type epitaxial layer,

wherein said optoelectronic stack is disposed between said first-type epitaxial layer and said second-type epitaxial layer,

wherein said first-type epitaxial layer is disposed on said substrate, and

wherein said first electrode is disposed on and contacts said first-type epitaxial layer, and said second electrode is disposed on and contacts said second-type epitaxial layer.

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